Difficult terms

1. Virus host prediction

This means predicting which organism (host) — for example, a human, bat, pig, or plant — a virus can infect.
Scientists use computer models to predict this from the virus's genetic information.

Example:
👉 "Does this virus infect humans or bats?"
👉 "Can this virus jump from animals to people?"

2. Using machine learning

They are using artificial intelligence (AI) algorithms to make those predictions automatically by learning patterns from known data.

Example:
The machine learns from many viruses where the host is already known, and then tries to predict the host of a *new* virus.

3. Short sequence k-mers

A k-mer is a short piece of DNA or RNA sequence of length *k* (like 4, 5, or 6 letters long).
Example: if k=3 and the sequence is `ATCGT`, then the 3-mers are: `ATC`, `TCG`, `CGT`.

Machine learning models use the frequency of these small sequence fragments (k-mers) to find patterns that differ between viruses that infect different hosts.

4. Taxonomic features

"Taxonomic" means related to classification — like species, genus, family, etc.
So "taxonomic features" refer to biological classification data about the virus or the host that can influence prediction.

Example:
A virus from the *Coronaviridae* family is more likely to infect mammals than insects — that's a taxonomic feature.

5. Host-dependent features

These are characteristics that depend on the host — for instance, the host's immune system, cell receptors, or typical environment — which can affect whether a virus can infect it.

6. Sample bias

This means that the data used for training the machine learning model might not be balanced or fully representative.
For example:

- If most data comes from human viruses, the model may be biased toward predicting human hosts.

- This can make predictions less accurate for other hosts like birds or reptiles.

7. "Train–Test Split" Means

In any machine learning (ML) experiment, the **dataset** (in this case, RNA virus genomes) is divided into:

- **Training set** → used to teach the AI model (the ML algorithm).

- **Test set** → used to check how well the trained AI performs on **new, unseen data**.

This separation is critical to measure how well the model generalizes to new viruses, not just memorizes old ones.

8. Baseline method

A baseline method is a simple, initial model used as a benchmark to evaluate and compare the performance of more complex models.

9. Host Taxon Predictor (HTP)

A previous ML-based host prediction tool published earlier (Galan et al., *Scientific Reports*, 2019).

How does "The effect of taxonomic, host- dependent features and sample bias on virus host prediction using machine learning and short sequence k-mers" actually work

Meaning:

Researchers studied how:

1. **Taxonomic features** (virus classification groups),

2. **Host-related genome patterns**, and

3. **Data imbalance (sample bias)**
   affect how well **AI models (machine learning)** can **predict which host a virus infects** by using **short pieces of its genetic code (k-mers)**.

- **Introduction (What & Why)**

- **New viruses are being discovered** all the time in "metavirome" studies (sequencing all viral DNA/RNA in a sample like animal feces or water).

- These samples contain **mixed genetic material** — from the host, food, or environment.

- To identify which virus infects which host, we need **fast, automatic methods** — that's where **machine learning** helps.

- Previous studies showed:

  - **Virus taxonomy** (its family/genus) and **short sequence patterns (k-mers)** can predict hosts.

  - But results vary a lot because of **differences in dataset composition and bias**.

👉 **Goal of this study:**
Find out how dataset setup (taxonomic diversity, bias, fragment length, etc.) affects ML prediction quality.

🔹 Methods

1️⃣ Data Collection

- Collected **17,482 RNA virus genomes** infecting:

- Mammals 🐄,
- Insects 🐜,
- Plants 🌿.

- Taken from **Virus–Host Database (Virus–Host DB)**.
- After cleaning duplicates and removing viruses infecting multiple hosts (like arboviruses),
  ➤ Final dataset = **1,363 unique genomes** from **42 virus families**.

2️⃣ **Sequence Features (k-mers)**

- Every virus genome was turned into **numbers** based on **k-mer frequencies** (short sequences of length *k*).
- Tested:
  - **Nucleotide k-mers:** lengths 1 to 7 (A, C, G, T patterns)
  - **Amino acid k-mers:** lengths 1 to 3 (protein-level patterns)
- Purpose: See which combination of these gives the best prediction.

3️⃣ Train-Test Splits (How They Tested AI)

They divided data into **training** (67–78%) and **testing** (22–33%) sets in 3 ways:

Split Type	Description	Purpose
Closely Related	Training and test contain similar virus families	Easy case
Non-overlapping Genera	Test genera not seen in training	Medium difficulty
Non-overlapping Families	Test families not seen in training	Very hard case

- Each setup was repeated **10 times** with random splits to check reliability.
- They also simulated **short fragments** (400 or 800 nucleotides) to mimic real sequencing data.

4️⃣ Machine Learning Models Used

Used 4 algorithms:

1. **Random Forest (RF)** 🌳
2. **LightGBM (LGBM)** ⚡
3. **XGBoost (XGB)** 🔺

4. **Support Vector Classifier (SVC)** 📊

Measured success with **F1-score** (balance of accuracy and recall).

5️⃣ Baseline (Comparison Methods)

Compared their ML models to:

- **Dummy classifier:** always guesses the most common host (for reference).
- **tBLASTx:** similarity search method (traditional, slower, uses sequence alignment).
- **Host Taxon Predictor (HTP):** another published ML-based host predictor.

These three baselines were used to prove that their new ML models (using k-mer features and algorithms like SVM, Random Forest, XGBoost) were not only **faster** than tBLASTx but also **more accurate and generalizable** than previous ML tools like HTP — especially for **predicting hosts of previously unknown viruses**.

🔹 Results

🧠 Predicting Hosts of Known Viruses

- Even **simple 1-mer** models (A/T/G/C frequency) predicted hosts quite well — F1 = 0.78–0.80.
- **4-mer + SVC** was best → F1 = 0.96.
- Better than Host Taxon Predictor (0.89) and similar to tBLASTx (0.93).
- So, ML can **match BLAST performance but much faster**.

🧠 **Predicting Hosts of *New* Genera or Families**

- Accuracy dropped because the model couldn't rely on taxonomic similarity anymore.
- For **unknown genera:**
  - F1 = 0.79 (SVC with 4-mers)
  - Better than Host Taxon Predictor (0.72) and tBLASTx (0.68).
- For **unknown families:**
  - F1 = 0.50–0.53 (hardest case).
  - So, **still some predictive power**, but not very strong.

🧠 **Taxonomic Richness & Dataset Composition**

- More virus types (diversity) in the training data → slightly better results.
- Some virus families were easier or harder to predict:
  - **Easy:** Picornaviridae (mammals), ssRNA(+) viruses.
  - **Hard:** Rhabdoviridae (infects many hosts), Bromoviridae (plants).

🧠 **Nucleotide vs Amino Acid Features**

- **Nucleotide (NT) features** were better than amino acid (AA) features.
- Combining both did not improve results.
- **AA k-mers** only helped for a few negative-sense RNA viruses (Paramyxoviridae, Rhabdoviridae).

🧠 **Predicting Hosts from Short Fragments (Realistic Case)**

- When predicting from **400–800 nt fragments**, accuracy dropped:
  - F1 ≈ 0.56–0.63

    - Best when models were trained on **fragments of the same length**.
    - Training on full genomes → worse results.
- **Best k-mer sizes:** 2–4.
- **tBLASTx** failed completely on short fragments.

🔹 Discussion (Meaning of the Findings)

1. **Two influences on host prediction:**
    - **Taxonomic signal** (similar viruses have similar patterns)
    - **Host-dependent features** (genome changes adapted to host biology)
2. **Key discovery:**
    - Dataset setup (composition & bias) affects performance **more** than the ML model choice.
    - **Short k-mers (2–4)** are enough for meaningful prediction.
    - Longer k-mers (>4) become too rare, making models unstable.
3. **Training on data similar to target tasks is essential.**

- For predicting hosts of short sequences → train on short sequences.
- For predicting new families → include diverse training viruses.

4. **Best algorithms:**
   - **SVC** = great for full genomes.
   - **Tree-based models (XGB, LGBM)** = stable for fragments.
   - No single universal best model.
5. **Limitations:**
   - Only tested on known virus sequences, not mixed metagenomic data.
   - Unbalanced datasets (more mammal viruses than insect/plant) can cause bias.

🔹 Conclusion (Simplified Summary)

- Machine learning can accurately **predict which host a virus infects** using just **short genome fragments**.
- **Short k-mers (2–4 nt)** contain enough information for this task.

- **SVC with 4-mers** achieved the best accuracy (F1 = 0.79) on unseen virus genera.

- Dataset **composition and balance** matter more than the specific ML algorithm.

- Traditional **BLAST** works well for known viruses but **fails for new ones**, while ML still works moderately.

- Useful for **biosafety and rapid virus identification** in metagenomic studies.

DeepHost: Fine grained viral host prediction and zoonotic potential via multitask deep learning

PREVIOUS PAPER

- Predicts **virus host group** (mammal, insect, plant) using **machine learning on k-mers**.

- Evaluates accuracy and bias, but only at the **broad host level** (no species, no zoonosis).

- Uses **classical ML**, not deep learning.

- Does **single-task classification** (only predicts host).

NEW IDEA

1. Predicts **specific hosts** — e.g., *bat, rodent, mosquito, rice* — instead of just broad categories.
   ➤ That's **finer taxonomic resolution**.

2. Predicts **zoonotic potential** — i.e., how likely it is for the virus to infect humans.
   ➤ That's a **completely new prediction dimension**.

3. Uses **deep learning**, which can handle multitask outputs.

4. Emphasizes **interpretability** — finding *why* the model predicts something.

5. Tests **generalization** on *new virus genera/families*, same as the paper — so it builds directly upon their experimental setup.

We can literally position our thesis as:

> "Building on [Author, 2025], this work extends virus-host prediction from classical ML to a deep multitask framework that simultaneously predicts fine-grained host taxonomy and zoonotic potential."

🔸 Your Extended Pipeline

Step	Description	How it Extends
1. Data Collection	Use Virus–Host DB + literature data on zoonotic viruses	Adds zoonotic labels
2. Feature Extraction	Use raw sequences or embeddings (DNABERT / CNN)	Replaces manual k-mers
3. Model Design	Deep multitask neural network	Learns shared viral features for both tasks
4. Task 1	Predict host taxa (species/genus/family levels)	Extends classification granularity
5. Task 2	Predict zoonotic potential (probability 0–1)	Adds new output dimension
6. Evaluation	Accuracy, F1, AUC, calibration, interpretability	Deeper evaluation
7. Generalization Test	Hold-out new virus families	Keeps same strong evaluation concept

🧩 OVERVIEW — What your research will do

You will:

1. Collect viral genomic sequences and their known hosts.
2. Add labels for zoonotic potential (whether they can infect humans).
3. Preprocess and encode the sequences numerically.

4. Build a **multitask deep learning model**:
    - Task 1: Predict host taxa (bat, rodent, mosquito, etc.)
    - Task 2: Predict zoonotic probability (infection risk to humans).
5. Train, evaluate, and interpret the model.

🧩 STEP-BY-STEP PROJECT PLAN

(with exact tasks + required knowledge per step)

🧬 STEP 1: Literature Review & Planning

🎯 **Goal:**

Understand what has been done and how your work extends it.

🧠 **Learn & Review:**

- Read the 2025 paper (your base).
- Study related papers on zoonotic prediction (Grange et al., 2021; Mollentze & Streicker, 2021).
- Learn basics of **virus taxonomy** and **zoonosis**.

- Identify data sources: *Virus–Host DB, NCBI Virus, VIRION, Zoonotic Virus DB*.

💡 **Deliverables:**

- Write background section in thesis.
- Define research gap and objectives.
- Formulate research questions and hypothesis.

⚙️ **Tools:**

- Google Scholar, Zotero (for citation management)

🧩 STEP 2: Data Collection

🎯 **Goal:**

Collect viral genomes + host taxonomy + zoonotic labels.

🧠 **Learn/Use:**

- **Bioinformatics databases:**
  Virus–Host DB, NCBI Virus, VIRION, Grange's dataset.
- **File formats:** FASTA (for sequences), CSV/TSV (for metadata)

- **APIs & scraping:** Python's `requests`, `Bio.Entrez` (Biopython)

⚙️ **Steps:**

1. Download RNA virus genomes (FASTA).
2. Extract metadata (host name, family, etc.).
3. Label zoonotic viruses (1 = infects humans, 0 = non-zoonotic).
4. Clean dataset:
    - Remove duplicates
    - Remove ambiguous or multi-host entries
    - Ensure balanced host types

📘 **Concepts to Know:**

- Genome sequences (A, C, G, U)
- FASTA format
- Basic data cleaning (Pandas)

🔢 STEP 3: Data Preprocessing and Encoding

🎯 **Goal:**

Convert raw genomic sequences into numerical tensors for deep learning.

🧠 **Learn:**

- **k-mer encoding** (for baseline ML)
- **One-hot encoding / embedding** (for DL input)
- **Sequence padding/truncation** (for fixed-length input)
- **Label encoding** (for host types)

⚙️ **Steps:**

1. Normalize genome lengths (e.g., take first 1,000–2,000 nucleotides or fragments).
2. Convert sequences into:
    - One-hot arrays (`A=1000`, `C=0100`, `G=0010`, `U=0001`)
    - or k-mer tokenized sequences (for Transformers)
3. Create one-hot labels for:
    - Host class (bat, rodent, mosquito, rice)
4. Create binary labels for:

    - Zoonotic potential (1/0)

🧮 **Tools/Libraries:**

- **Python, Biopython**
- **NumPy / Pandas**
- **Scikit-learn** (for preprocessing)

🤖 STEP 4: Build Deep Learning Model (Core Part)

🎯 **Goal:**

Build a **multitask neural network** that predicts host + zoonosis.

⚙️ **MODEL ARCHITECTURE**

🧩 **Input:**

- One-hot or embedded viral genome sequence
- Shape: `(sequence_length, 4)` or `(sequence_length, embedding_dim)`

🧩 **Shared feature extractor:**

A shared neural layer learns general viral patterns useful for both tasks.

Example (CNN-based):

```
Conv1D(128, kernel_size=7, activation='relu')
MaxPooling1D(pool_size=3)
Dropout(0.3)
Conv1D(64, kernel_size=5, activation='relu')
GlobalMaxPooling1D()
Dense(256, activation='relu')
```

🧩 **Output Heads:**

Head 1 (Host Taxonomy Prediction):

```
Dense(num_host_classes, activation='softmax')
```

-

Head 2 (Zoonotic Potential Prediction):

```
Dense(1, activation='sigmoid')
```

-

Both share the same backbone network.

🧩 **Combined Model:**

```
Model(inputs=input_seq, outputs=[host_output,
zoonotic_output])
```

⚙️ **Loss Function (Multitask)**

Combine classification + probability tasks:

Ltotal=α×Lhost+β×LzoonoticL_{total} = \alpha \times L_{host} + \beta \times L_{zoonotic}Ltotal=α×Lhost+β×Lzoonotic

where

- LhostL_{host}Lhost = categorical cross-entropy
- LzoonoticL_{zoonotic}Lzoonotic = binary cross-entropy
- α,β\alpha, \betaα,β = weights (e.g., 0.7, 0.3)

🧠 **Learn:**

- Neural network basics: layers, activation functions, loss, optimizer
- **Keras or PyTorch**
- **Multitask learning** principles
- Model training/validation/test splits

⚙️ STEP 5: Training the Model

🎯 **Goal:**

Train on known viruses, test on new genera/families.

🧠 **Learn:**

- Train/test/validation splitting (stratified)
- Model callbacks (early stopping, checkpoint)
- Overfitting control (dropout, regularization)

⚙️ **Steps:**

1. Split dataset (70% train, 15% val, 15% test).
2. Train model for 20–50 epochs.
3. Use GPU (Google Colab or Kaggle).
4. Save best model (`model.save()`).

📊 STEP 6: Evaluation

🎯 **Goal:**

Measure how well model generalizes.

🧮 **Metrics:**

Task	Metrics
Host taxonomy	Accuracy, F1-score (macro & weighted)

Zoonotic potential	AUC, Precision, Recall, F1
Overall	Multitask loss, confusion matrices

🧠 **Learn:**

- Classification metrics (precision, recall, F1)
- ROC-AUC curve
- Calibration curves (for zoonotic probabilities)
- Confusion matrix plotting (Seaborn, Matplotlib)

🔍 STEP 7: Interpretability & Visualization

🎯 **Goal:**

Make the model explainable.

🧠 **Learn:**

- **SHAP values** (feature importance)
- **Attention maps** (for Transformer models)
- **t-SNE / PCA** (visualizing embeddings)

- Model interpretability techniques

⚙️ **Output:**

- Visualize which genomic regions most influence prediction.
- Highlight k-mers or motifs correlated with zoonosis.

🧩 STEP 8: Comparative Analysis

🎯 **Goal:**

Show that your deep model outperforms old ML methods.

⚙️ **Do:**

- Implement ML baselines: SVM, Random Forest, XGBoost using k-mer frequencies.
- Compare metrics with your deep multitask model.
- Include table and graphs.

🧠 **Learn:**

- scikit-learn ML basics
- Evaluation interpretation

🧩 STEP 9: Write Thesis / Research Paper

Structure:

Section	Description
Abstract	200–250 words summary of purpose, methods, results
Introduction	Background, problem statement, motivation
Literature Review	Related works (virus-host prediction, zoonosis models)
Methodology	Dataset, encoding, model design, multitask setup
Results	Accuracy tables, graphs, confusion matrices
Discussion	Interpret what worked, what didn't
Conclusion	Key findings + future work
References	Cite 2025 paper, zoonosis studies, DL papers

🧩 STEP 10: Tools and Environment Setup

Category	Tool / Library	Learn
Coding	**Python (3.8+)**	syntax, file I/O

Data Handling	**Pandas, NumPy**	dataframes, arrays
Bioinformatics	**Biopython**	FASTA reading, parsing
ML	**Scikit-learn**	classical models
DL	**TensorFlow / Keras** or **PyTorch**	model design, training
Visualization	**Matplotlib, Seaborn**	plots, heatmaps
Interpretability	**SHAP**, **LIME**	explain model
Deployment (optional)	**Streamlit**	web demo UI

🧠 KNOWLEDGE MAP — What You'll Need to Learn

Stage	Concept/Skill	Time Needed
Data Prep	FASTA files, Biopython, Pandas	1 week
Encoding	k-mer, one-hot, padding	2–3 days
DL Basics	Layers, activations, loss functions	1 week

CNN/Transformer	Model design and training	2–3 weeks
Multitask Learning	Shared backbone, multiple outputs	1 week
Evaluation	F1, AUC, confusion matrix	3–5 days
Interpretability	SHAP, feature importance	1 week
Writing & Visualization	Report structure, plots	1–2 weeks

So roughly **8–10 weeks total** for a strong thesis-level project.

🧩 📘 Example Research Objectives (for your thesis proposal)

1. **To develop** a multitask deep learning model that predicts viral host taxa and zoonotic potential directly from genomic sequences.

2. **To compare** the performance of deep learning models against classical ML-based k-mer approaches.

3. **To evaluate** the model's ability to generalize to novel viral genera and families.

4. **To interpret** learned sequence features to identify host- and zoonosis-associated genomic patterns.